

# Types of Computers

Lesson 02 — Classify by Signal and Scale

JKSSB Computer Awareness

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## Video Lesson 02

Watch: Computer Classifications for JKSSB

# One Device, More Than One Classification

- ▶ A laptop is a personal computer by intended use and a digital computer by the way it represents data.
- ▶ These labels answer different questions.

## Classification rule

First identify the basis: signal handled, size/capability, intended purpose, or form factor.

# By the Kind of Data Signal

## Analog

- ▶ Represents continuously varying quantities.
- ▶ Example: a traditional analog speedometer.

## Digital

- ▶ Represents data as discrete values, commonly binary digits.
- ▶ Example: a laptop.

## Hybrid

- ▶ Combines analog input or measurement with digital processing.
- ▶ Example: a digital patient monitor.

# Analog, Digital, Hybrid: The Contrast

- ▶ **Analog**: values vary continuously over a range.
- ▶ **Digital**: values are represented as distinct symbols or states.
- ▶ **Hybrid**: a system uses both forms in different stages.

## Common trap

A digital computer can process a measurement that began as an analog signal; that does not make the whole computer an analog computer.

## By Traditional Size and Capability

- ▶ **Microcomputer**: personal-use system, such as a desktop, laptop, tablet, or smartphone.
- ▶ **Minicomputer**: traditional mid-range, multi-user class; the label is historical and less common in modern product marketing.
- ▶ **Mainframe**: high-volume transaction and input/output workloads for many users.
- ▶ **Supercomputer**: very high-performance parallel computation for large scientific or engineering workloads.

# Mainframe Is Not Supercomputer

## Mainframe

- ▶ Many concurrent users or transactions
- ▶ Reliability and throughput
- ▶ Example workload: large banking records

## Supercomputer

- ▶ Huge calculations that can be parallelized
- ▶ High computational performance
- ▶ Example workload: climate simulation

Do not decide by physical size alone; classify by the system's characteristic workload.

## By Purpose and Form

- ▶ **General-purpose**: runs many kinds of programs, such as a personal computer.
- ▶ **Special-purpose**: designed for a narrower task, such as an embedded controller.
- ▶ **Embedded computer**: built into a larger device to control or support its function.
- ▶ Desktop, laptop, tablet, and smartphone describe common form factors, not analog/digital signal classes.

## Quick Matching

- ▶ Continuous temperature measurement: **analog signal**.
- ▶ Smartphone: **digital microcomputer**.
- ▶ Many bank transactions at once: **mainframe workload**.
- ▶ Parallel weather-model calculation: **supercomputer workload**.
- ▶ Washing-machine controller: **embedded, special-purpose**.

## Generation Link: Keep the Axes Separate

- ▶ Computer **type** can mean its signal class, scale, purpose, or form.
- ▶ Computer **generation** refers to historical technology eras.
- ▶ The next history lesson connects vacuum tubes, transistors, integrated circuits, and microprocessors to generations.

### Exam cue

Integrated circuits are associated with the third generation; the microprocessor is associated with the fourth.

## Lesson 02: Recall

- ▶ Analog = continuous; digital = discrete; hybrid = both.
- ▶ Microcomputer = personal-use category; embedded = built into a larger product.
- ▶ Mainframe emphasizes throughput and many users; supercomputer emphasizes high-performance computation.
- ▶ Name the basis of classification before selecting an answer.